Chest-Xray

1. Introduction

Pneumonia is a serious respiratory infection that affects the lungs and can become life-threatening if not diagnosed early. Traditional diagnosis relies on medical experts analyzing chest X-ray images, which can be time-consuming and sometimes subjective depending on experience.

With the advancement of Artificial Intelligence (AI) and Deep Learning, it has become possible to build automated systems that assist in medical image analysis with high accuracy. These systems help reduce diagnostic time and improve consistency in predictions.

In this project, we developed an AI-based Pneumonia Detection System using **Deep Learning and Transfer Learning techniques**. The system analyzes chest X-ray images and classifies them into two categories:

- NORMAL
- PNEUMONIA

The model is trained using a large dataset of chest X-ray images and deployed using a web-based API built with **FastAPI**, along with a user-friendly web interface for real-time predictions.

The goal of this project is to build an efficient, accurate, and deployable AI system that can assist in early pneumonia detection.

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2. Dataset Description

The dataset used in this project is a publicly available chest X-ray dataset.

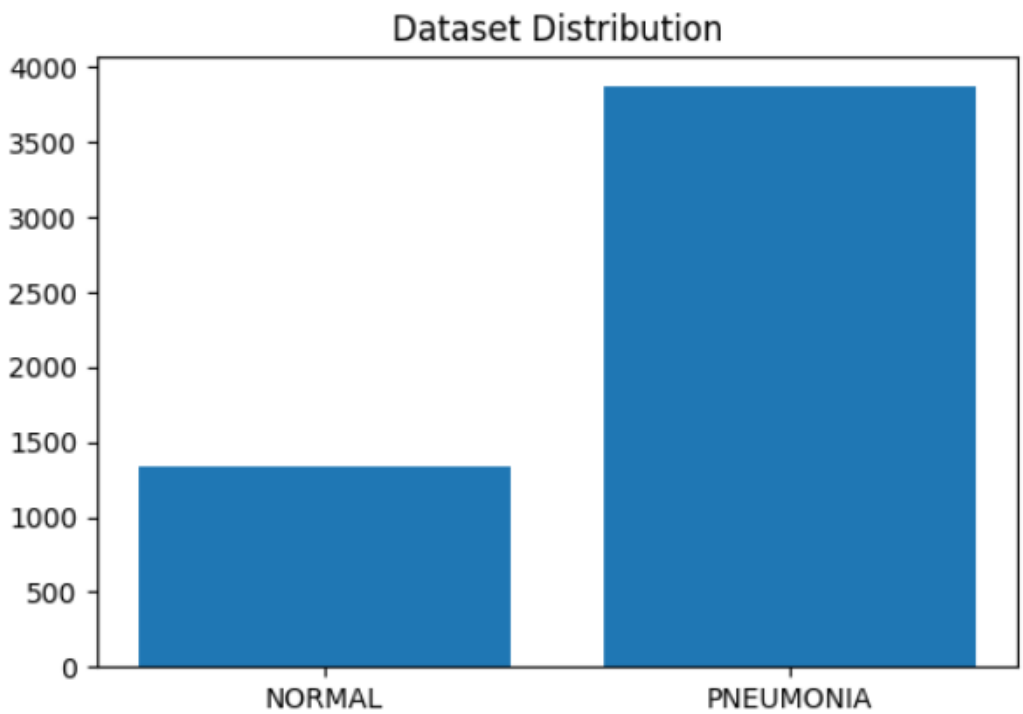
Dataset Details:

- Total images: **5856**
- Training set: **5216**

- Validation set: **16**
- Test set: **624**

Class Distribution:

- NORMAL: **1341 images**
- PNEUMONIA: **3875 images**




```
Classes: ['NORMAL', 'PNEUMONIA']
Counts: [1341, 3875]
```

Observations:

- The dataset is imbalanced (more pneumonia cases than normal).
- This imbalance is handled using Weighted Random Sampling during training.

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3. Data Preprocessing

Before training the model, several preprocessing steps were applied to improve performance and generalization.

Steps used:

- Resizing all images to **224 × 224**

- Converting images to 3 channels (RGB)
- Normalization using ImageNet mean and standard deviation
- Data augmentation for training set:
    - Random rotation
    - Horizontal flipping
    - Brightness/contrast adjustment
    - Affine transformations

Purpose of preprocessing:

- Improve model generalization
- Reduce overfitting
- Standardize input format for ResNet18

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4. Model Architecture

The model used in this project is **ResNet18**, a pre-trained convolutional neural network.

Why ResNet18?

- Strong performance in image classification
- Pre-trained on ImageNet dataset
- Efficient and lightweight compared to deeper models

Architecture:

Input Image → ResNet18 Feature Extractor → Fully Connected Layers → Softmax Output

Modifications:

- Final layer replaced with:
    - Linear layer (512 → 256)
    - ReLU activation
    - Dropout layers
    - Output layer (2 classes)

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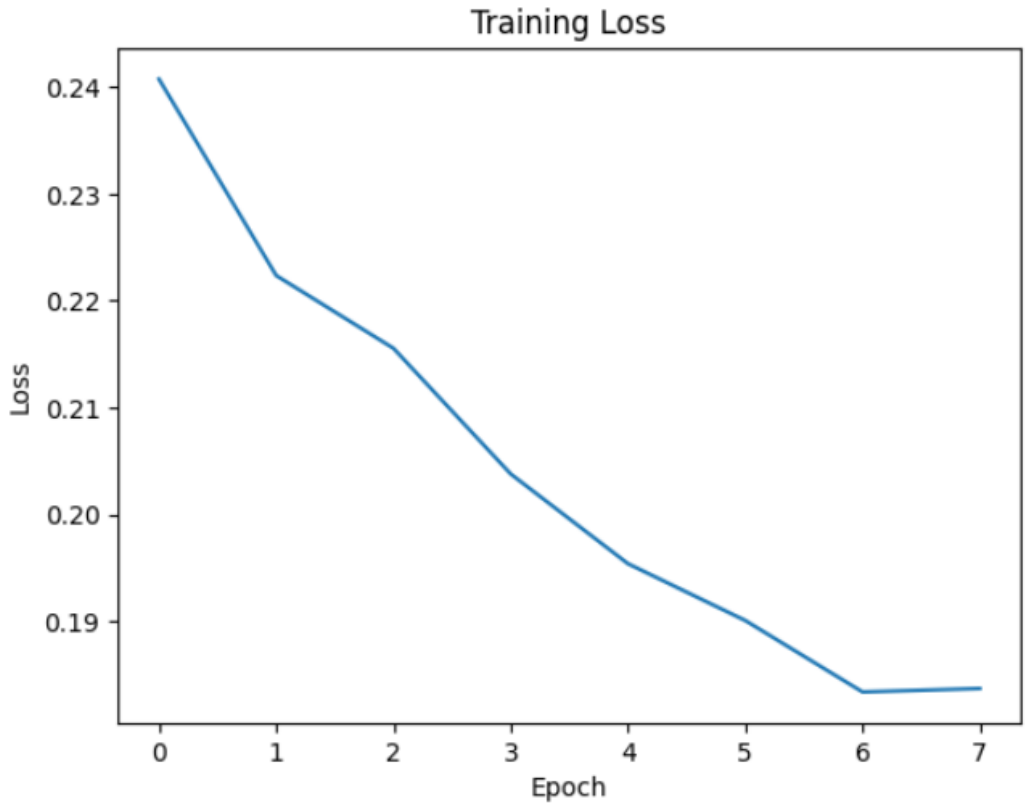
5. Training Configuration

Hyperparameters:

- Epochs: **10**
- Batch size: **32**
- Learning rate: **0.0003**
- Optimizer: Adam
- Loss function: CrossEntropyLoss

Techniques used:

- Transfer Learning
- Data Augmentation
- WeightedRandomSampler (to handle class imbalance)
- Early stopping (to prevent overfitting)



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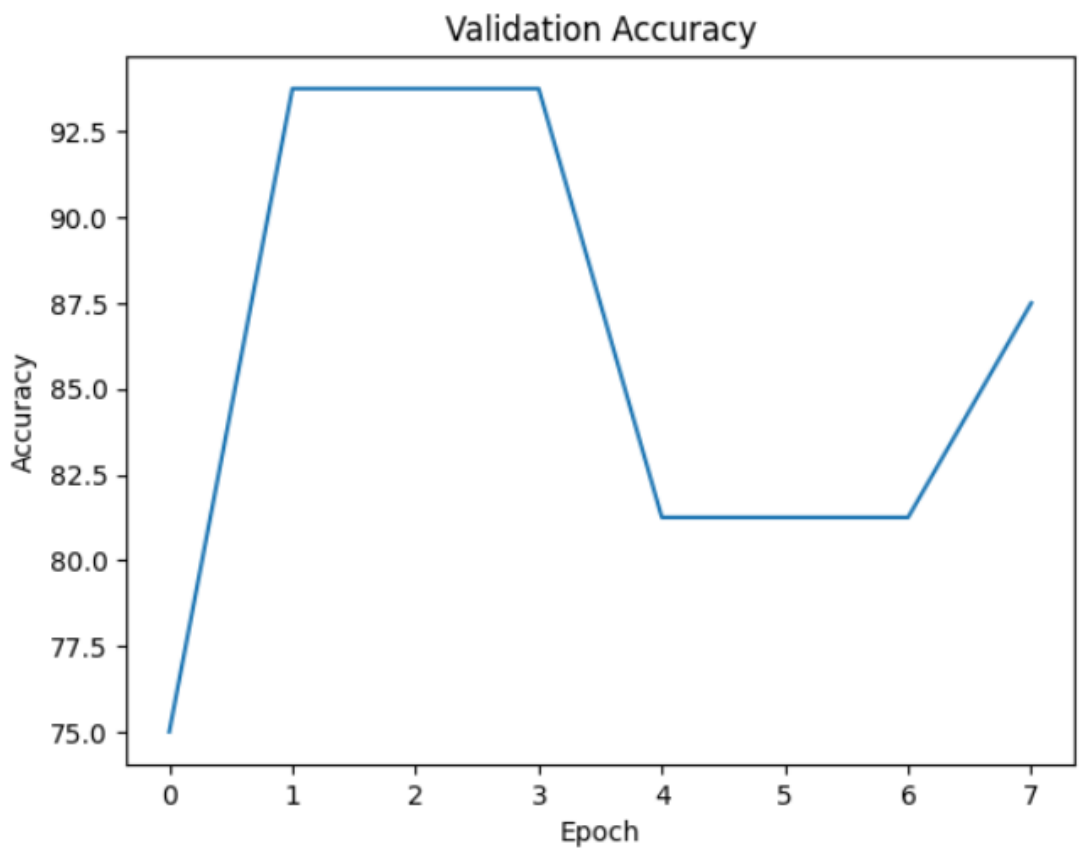
6. Validation Results

The validation dataset was used to monitor model performance during training.

Final Validation Performance:

- Accuracy: **~90%**
- Observed behavior: fluctuating but converging trend



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Explanation:

Fluctuations in accuracy are normal due to:

- Data augmentation randomness
- Small validation dataset
- Stochastic gradient descent behavior

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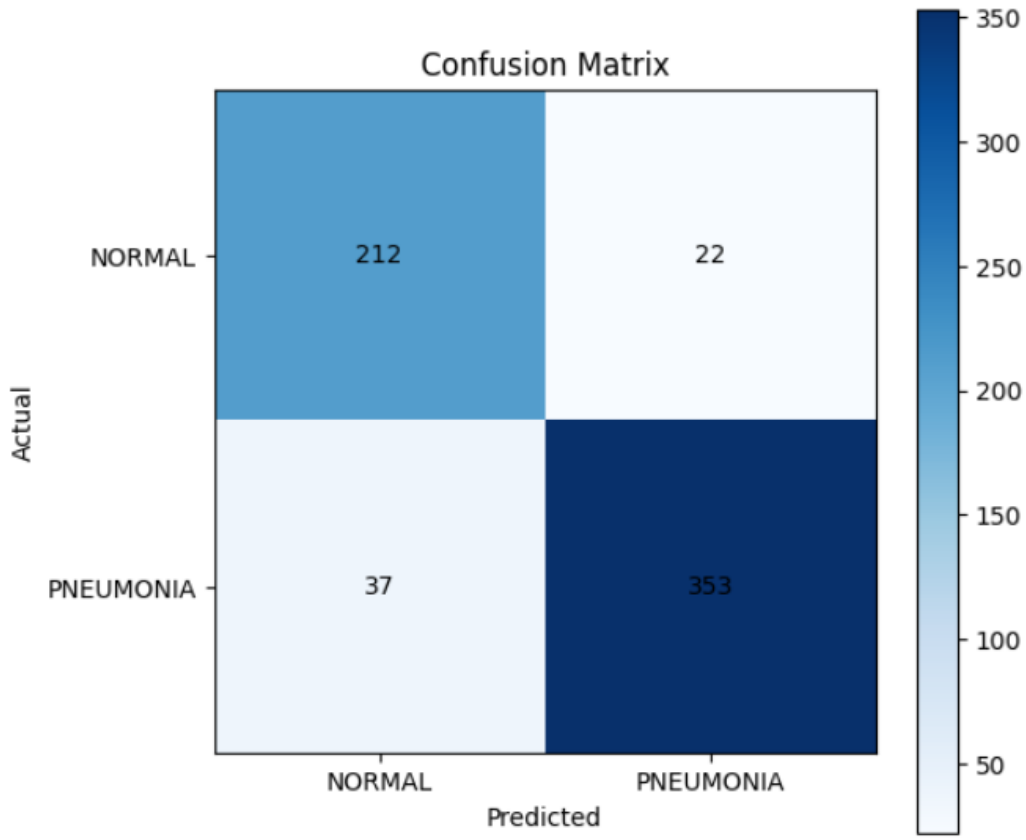
7. Test Results (Final Evaluation)

The test set was used only once after training to evaluate real-world performance.

Results:

- Test Accuracy: **0.90%**
- Precision (Pneumonia): **0.90**
- Recall (Pneumonia): **0.90**

- F1-Score: **0.91**
- ROC-AUC: **0.95**
- 

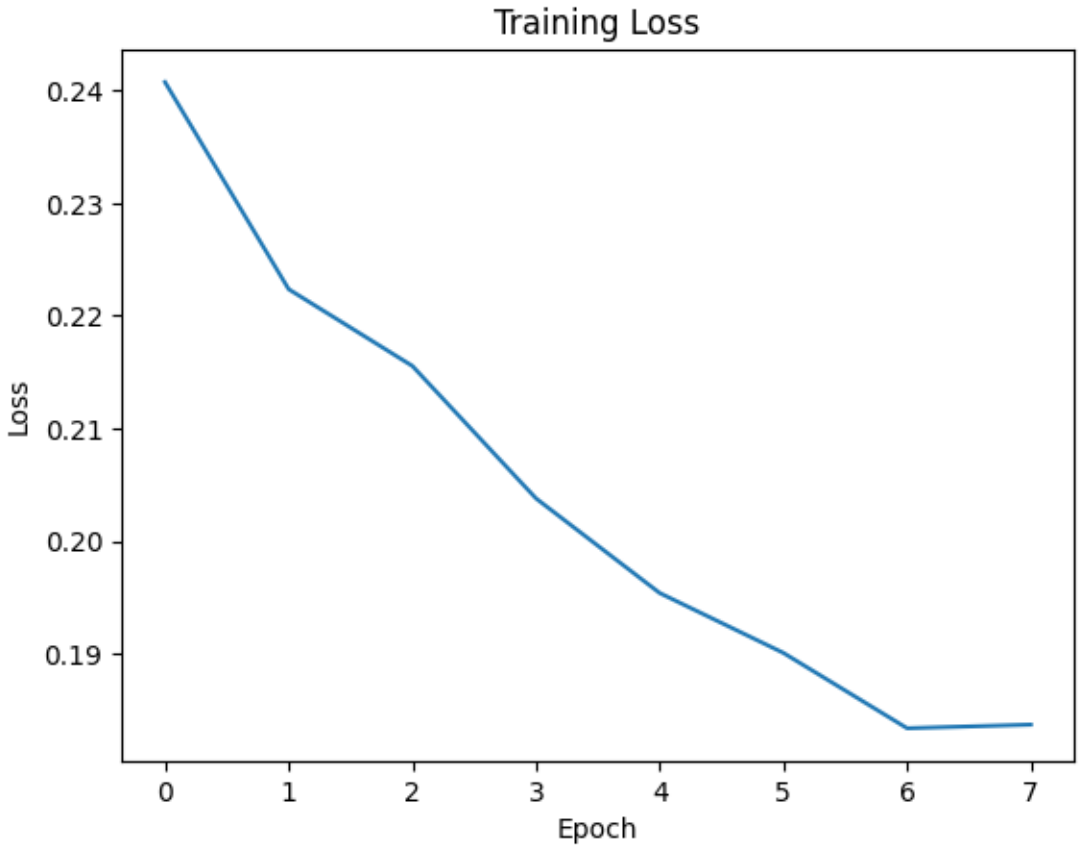

```python
print(classification_report(
    all_labels,
    all_preds,
    target_names=test_dataset.classes
))
```

```
              precision    recall  f1-score   support

      NORMAL       0.87      0.88      0.88       234
   PNEUMONIA       0.93      0.92      0.92       390

    accuracy                           0.91       624
   macro avg       0.90      0.90      0.90       624
weighted avg       0.91      0.91      0.91       624
```

CONFUSION MATRIX



Interpretation:

- High recall indicates strong ability to detect pneumonia cases.
- ROC-AUC shows excellent classification capability.

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8. API Development (FastAPI)

The trained model was deployed using FastAPI.

Features:

- /predict endpoint
- Image upload support
- JSON response output
- Model inference in real time

API Flow:

User Image → FastAPI → Preprocessing → ResNet18 Model → Prediction Output

Output format:

```
{
  "class": "PNEUMONIA",
  "confidence": 0.93
}
```

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9. User Interface (UI)

A simple web-based UI was created using HTML, CSS, and JavaScript.

Features:

- Upload chest X-ray image
- Preview image before prediction
- Display predicted class
- Show confidence score

Purpose:

- Make AI system usable for non-technical users
- Provide real-time interaction with model

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10. System Deployment

The system can be deployed using:

- FastAPI server
- Uvicorn
- Docker (optional)

Run command:

```
uvicorn app:app --reload
```

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11. Performance Summary

Metric	Value
Accuracy	~90%
Precision (Pneumonia)	0.90
Recall (Pneumonia)	0.90
F1-score	0.91
ROC-AUC	0.95

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12. Conclusion

In this project, we successfully developed a deep learning-based Pneumonia Detection System using ResNet18 and transfer learning techniques. The system achieved strong performance with approximately 90% accuracy and high ROC-AUC score of 0.95.

The integration of FastAPI and a web-based UI makes the system accessible and practical for real-world use. The model demonstrates the potential of AI in assisting medical diagnosis and improving healthcare efficiency.